

Kunal Maheshwari

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Summary

Software Engineer / Java Backend Engineer with 4+ years building and maintaining high-availability healthcare and automotive systems, including services handling 50K+ daily requests under a 99.99% SLA.

Delivery scope: Carries product and system changes from requirement clarification and component design through development, unit/integration testing, peer review, documentation, release validation, and production maintenance.

Engineering approach: Reasons across service, network, data, and business-rule boundaries; uses telemetry, failure-path analysis, and measured performance evidence to build reliable, maintainable, and efficient solutions.

Working style: Takes ownership of priorities and deliverables, communicates risks and trade-offs, collaborates with technical and product-facing stakeholders, and raises team quality through design discussions, code reviews, reusable documentation, and mentoring.

Technical Skills

Programming/Computer Science: Java 8/11, Core Java, OOP, Collections, Streams, data structures and algorithms, design patterns, system and component design; Python, TypeScript, JavaScript (projects).

Backend/Systems: Spring Framework/MVC, REST/SOAP APIs, Java EE/J2EE, Hibernate/HQL, JDBC/OJDBC, enterprise SOA and distributed-service integration, asynchronous messaging, networking and security principles, performance tuning, observability.

Data/Cloud: Oracle, DB2, PostgreSQL (project/training), SQL and data-access optimization; Azure Service Bus/Relay/Functions (professional), AWS S3/Lambda/EFS/ECS/RDS and Docker/Terraform (project).

Quality/Delivery: JUnit/JUnit4, integration testing, SoapUI, Postman, code and design reviews, debugging, root-cause analysis, Jenkins, Maven, IBM UrbanCode Deploy, GitLab CI/CD, Git, GitHub Actions, technical documentation, Agile/Scrum.

Work Experience

NTT Data, Bengaluru, India

Software Developer, Aug 2022 - Aug 2024

- Developed and maintained **Java 8/11, Spring Framework/JAX-WS, and REST/SOAP** healthcare payer services on IBM WebSphere, supporting **50K+ daily requests at 99.99% SLA availability**.
- Integrated Azure Relay, Azure Service Bus, Azure Functions, JMS, and synchronous downstream REST services with explicit timeout and partial-failure behavior across enterprise service flows.
- Integrated an AOP-based monitoring library that propagated correlation IDs and captured **latency, success/failure, and request-volume telemetry** for issue triage and service-quality analysis.
- Investigated a WebSphere memory/resource-retention incident by tracing code and service interactions, assessing operational impact, supporting mitigation, and validating rolling restarts across four load-balanced nodes.
- **Led a project-wide Log4j2 migration** prompted by security review, owning the PoC, dependency and configuration decisions, reusable guidance, code reviews, and coordinated rollout.
- Integrated Java services with Oracle through OJDBC, consuming established views and PL/SQL procedures while applying member/provider rules and validating downstream propagation.
- Coordinated requirements, implementation, JUnit/SoapUI testing, peer reviews, Jenkins/Maven/UrbanCode workflows, documentation, release checks, and go/no-go or rollback recommendations.

Capgemini, Mumbai, India

Senior Analyst, Jul 2020 - Aug 2022

- Developed and supported a Java EE/J2EE, JSF, Hibernate/HQL application on Oracle WebLogic with DB2 for Mercedes/Daimler Vans configuration, recommendation, and ordering workflows.
- **Reduced VIN repeat-configuration API latency from 60+ seconds to 10–20 seconds** through set-based HQL/DB2 mapping while retaining authoritative business validation in Java.

- Integrated a Python prediction service through asynchronous Java REST requests and callbacks, then applied deterministic option/package rules before recommendations reached dealer users.
- Triaged defects across Python output, Java validation logs, API behavior, and DB2 reference data; used Postman and production logs to isolate sources and verify corrections.
- Supported Jenkins/GitLab CI/CD, Git code reviews, JUnit4 testing, Agile delivery, and documentation; mentored junior developers through walkthroughs, debugging, and reviews.

Capgemini, Mumbai, India

Analyst Trainee, Jan 2020 - Jun 2020

- Built Java Spring/Spring MVC, Angular, and PostgreSQL training applications with authentication, email/OTP login, CRUD, balances, transactions, and profile workflows.

Selected Projects

CodingHelper - AI-Assisted Java/Python Learning Platform

- Built a React/TypeScript and FastAPI application using **LangChain**, **Ollama/Qwen 2.5 Coder**, **RAG**, **Qdrant**, and **PostgreSQL** to generate and retrieve structured Java/Python data-structure and algorithm solution ladders.
- Implemented isolated code execution and repair, rejection, and promotion gates; generated implementations must pass bounded asserting tests before display or reuse.

AWS Automated Spark Pipeline

GitHub

- Designed an event-driven S3-to-Lambda-to-EFS pipeline with a Docker-packaged Spark Structured Streaming workload on ECS and RDS PostgreSQL persistence; provisioned VPC/IAM and services through Terraform.
- Processed an **8GB+ dataset**, persisted checkpoints for restart recovery, and surfaced execution state through CloudWatch logging.

Interactive Portfolio System

- Built and deployed a React/TypeScript SPA with project case studies, searchable skill evidence, persisted theming, responsive UI, GitHub Actions checks, and GitHub Pages deep-link fallback.

Education

Central Michigan University, Mount Pleasant, MI, USA

Aug 2024 - Dec 2025

Master of Science in Computer Science, GPA: 3.87/4.0

Lovely Professional University, Punjab, India

Aug 2016 - May 2020

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.02/10

Certifications

Microsoft Certified: Azure Fundamentals | Azure Developer Associate | Azure Data Engineer Associate